In [1]:
```
# Import libraries
import pandas as pd

from sklearn.datasets import load_iris
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Load Iris dataset
iris = load_iris()

# Create DataFrame
df = pd.DataFrame(iris.data, columns=iris.feature_names)

# Independent variables (X)
X = df

# Dependent variable (y)
y = iris.target

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create Logistic Regression model
model = LogisticRegression(max_iter=200)

# Train the model
model.fit(X_train, y_train)

# Make predictions
y_pred = model.predict(X_test)

# Print results
print("Accuracy:", accuracy_score(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Predict a new flower
sample = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(sample)

print("\nPredicted Class:", prediction[0])
print("Predicted Species:", iris.target_names[prediction][0])
```

```
Accuracy: 1.0

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        10
           1       1.00      1.00      1.00         9
           2       1.00      1.00      1.00        11

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30


Predicted Class: 0
Predicted Species: setosa
C:\Users\admin\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2739: UserWar
ning: X does not have valid feature names, but LogisticRegression was fitted with fe
ature names
  warnings.warn(
```

In [ ]: